

Technology Stack

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Tailwind CSS	Provides a highly responsive single-page user interface to view real-time flood heatmaps, predictive analysis graphs, and disaster alert logs dynamically without page reloads.

2	Backend / Server-Side	Node.js (Express), Python (FastAPI)	Node.js handles high-volume simultaneous notification relays and user requests efficiently, while FastAPI rapidly executes our Python machine learning flood prediction models.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MongoDB, PostgreSQL	MongoDB handles unstructured, high-frequency live environmental IoT sensor log streams seamlessly, while PostgreSQL safely manages user accounts, roles, and historical records.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	AWS (Amazon Web Services), Docker	Docker containers ensure identical staging environments across setups, and AWS provides highly scalable cloud servers to deploy backend APIs and handle heavy seasonal traffic spikes.
5	Version Control & CI/CD	GitHub, GitHub Actions	GitHub maintains our source code repositories cleanly with continuous team tracking, and GitHub Actions automates test pipelines and cloud deployments whenever we update code.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenWeatherMap API, Twilio API	OpenWeatherMap API pulls precise localized live precipitation and storm cloud metrics, and Twilio API dispatches critical emergency evacuation SMS alerts instantly to citizens.